Egypt Smart City Analyzer
An End-to-End Data Engineering & AI Platform for Urban Intelligence

1. Project Description

This project involves the development of a comprehensive, end-to-end data engineering platform that transforms raw urban data into actionable business intelligence. The system collects, streams, processes, warehouses, and models city-level data from multiple sources, then exposes the results through an AI-powered interactive web dashboard.

The platform is designed to scale across any Egyptian city. Alexandria serves as the pilot city — a proof of concept that demonstrates the full pipeline from raw data ingestion to live AI-driven recommendations for commercial property suitability.

Data is collected from five domains: rental prices (Aqarmap), business locations (OpenStreetMap + Egyfinder), educational facilities (OSM), population density (CAPMAS + WorldPop + Google Earth Engine), and traffic networks (OSMnx). Data is streamed via Apache Kafka into a PostgreSQL Bronze layer, transformed by Apache Spark into a Gold layer, and warehoused in a SQL Server Star Schema. Two ML models are trained on the warehouse data and served via a FastAPI REST API, consumed by a Next.js web dashboard deployed on Vercel.

2. Project Team & Role Distribution

- **Nora Alaa Mahmoud Eldehimy:** Team Leader & Data Ingestion Engineer
    - Responsibilities included leading and coordinating the project team, ensuring milestones were met, and designing the overall pipeline architecture. Responsibilities also included building Kafka producers and consumers for real-time data streaming and overseeing data loading into the PostgreSQL Bronze layer.
- **Fatma Maher Fayed:** Data Ingestion & Pipeline Developer
    - Responsibilities included developing Kafka pipeline scripts for rental and business data collection. The role also involved integrating data into PostgreSQL and validating ingested data schema and completeness across all pipeline runs.
- **Menna Youssef Abd Alaziz Youssef:** Data Processing & ETL Engineer
    - Responsibilities included designing and implementing the PySpark-based ETL pipeline using the Medallion Architecture. This involved computing Haversine distances, calculating commercial suitability scores, and producing the structured Gold layer Dim and Fact CSV files.
- **Alzahraa Mohamed Fahmy:** Orchestration & Pipeline Automation Engineer
    - Responsibilities included designing and implementing the Apache Airflow DAG to orchestrate all pipeline phases in correct dependency order. The role also involved configuring Docker services for Airflow and scheduling the pipeline to run automatically every 6 months.
- **Alzahraa Mohamed Sami Mohamed:** Data Warehouse & AI Engineer

    - Responsibilities included designing and populating the SQL Server Star Schema Data Warehouse with 3 dimension tables and 2 fact tables, applying PKs, FKs, and indexes. The role also involved training and evaluating the RandomForestClassifier and GradientBoostingRegressor models, and exposing predictions through a FastAPI REST API.
- **Maryam Waheed Ahmed Rakha:** Frontend Developer & Dashboard Engineer
    - Responsibilities included building the Next.js interactive web dashboard with filters, map overlays, charts, and an AI recommendation panel. The role also involved deploying the frontend to Vercel and integrating it with the FastAPI backend.

3. Objectives

1. To deploy and manage 5 real-time Kafka pipelines that collect and stream urban data (rental prices, businesses, education, population, and traffic) into a PostgreSQL Bronze layer.
2. To implement a PySpark-based ETL pipeline using the Medallion Architecture to clean, enrich, and transform Bronze data into a structured Gold layer with computed suitability scores and Haversine-based distance features.
3. To design and populate a SQL Server Star Schema Data Warehouse comprising 3 dimension tables and 2 fact tables (3,548 rows total), enforcing referential integrity through primary keys, foreign keys, and indexes.
4. To orchestrate the full data pipeline using Apache Airflow, with a single DAG managing all phases in correct dependency order on a scheduled 6-month interval.
5. To train and evaluate two ML models — a RandomForestClassifier for commercial property tier classification (High / Medium / Low) and a GradientBoostingRegressor for continuous suitability scoring (1–10) — and expose them via a FastAPI REST API.
6. To develop and deploy a responsive Next.js web dashboard that enables entrepreneurs and investors to query urban analytics, visualize AI-driven recommendations, and export results as CSV for further analysis in Power BI or Excel.

4. Tools & Technologies

- **Data Ingestion & Streaming:** Apache Kafka, Python (kafka-python, BeautifulSoup4, Requests, Pandas, NumPy).
- **Bronze Storage:** PostgreSQL 16.
- **Data Processing (ETL):** Apache Spark (PySpark) 3.5.3, Pandas, NumPy.
- **Data Warehouse:** SQL Server, Star Schema (3 Dimensions, 2 Facts).
- **Orchestration:** Apache Airflow 2.9, Docker, Docker Compose.
- **Machine Learning & API:** scikit-learn (RandomForestClassifier, GradientBoostingRegressor), FastAPI, Uvicorn.
- **Frontend & Deployment:** Next.js, TypeScript, Tailwind CSS, Vercel (frontend), Render (backend).

- **Data Sources:** OpenStreetMap (Overpass API), Aqarmap, Egyfinder, CAPMAS, WorldPop, Google Earth Engine.
- **Version Control:** Git & GitHub.

5. KPIs (Key Performance Indicators)

7. Data Ingestion & Streaming (Kafka + PostgreSQL):
    a. 100% of scraped records successfully published to Kafka topics without data loss.
    b. End-to-end latency from scraping to PostgreSQL Bronze storage maintained under 2 seconds per batch.
    c. All 5 pipelines deliver records with correct schema and data types.
8. ETL Processing (Apache Spark):
    d. All Bronze records successfully transformed and merged into Gold layer CSV files.
    e. Haversine distance computation accuracy within ±5 meters for competitor proximity features.
    f. Suitability score distribution covers all three tiers (High, Medium, Low) across the Fact table.
9. Data Warehouse (SQL Server Star Schema):
    g. Star Schema fully populated: Dim_Business_Type (12), Dim_Area (51), Dim_Property (221), Fact_Area_Business_Score (612), Fact_Property_Suitability (2,652) — 3,548 total rows.
    h. 100% referential integrity enforced with zero constraint violations.
    i. Analytical query execution time under 3 seconds on standard hardware.
10. Orchestration (Apache Airflow):
    j. Full pipeline DAG executes all phases in correct dependency order with 0 failed tasks on end-to-end test run.
    k. DAG schedule triggers the pipeline at 2:00 AM on the 1st of every 6 months.
11. AI Models & API (scikit-learn + FastAPI):
    l. Tier Classifier achieves accuracy ≥ 85% via 5-fold stratified cross-validation.
    m. Score Regressor achieves $R^2 \geq 0.80$ on held-out validation data.
    n. API response time for /predict and /recommend endpoints under 500 ms per request.
12. Web Dashboard (Next.js + Vercel):
    o. Dashboard loads all district data within 2 seconds on standard broadband connection.
    p. All 12 districts and 6 business types queryable with correct AI score results returned.
    q. CSV export produces correctly formatted files compatible with Power BI and Excel.
    r. Dashboard data refresh latency from API under 2 seconds from user interaction.